



Tamarly

Research and advisory

TAMARLY MAGAZINE
ISSUE 001, SEPTEMBER 2026

The Rebuild

How artificial intelligence is being financed, fought over, regulated and lived with, and what it is doing to the shape of the century.

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TAMARLY MAGAZINE, ISSUE 001, SEPTEMBER 2026

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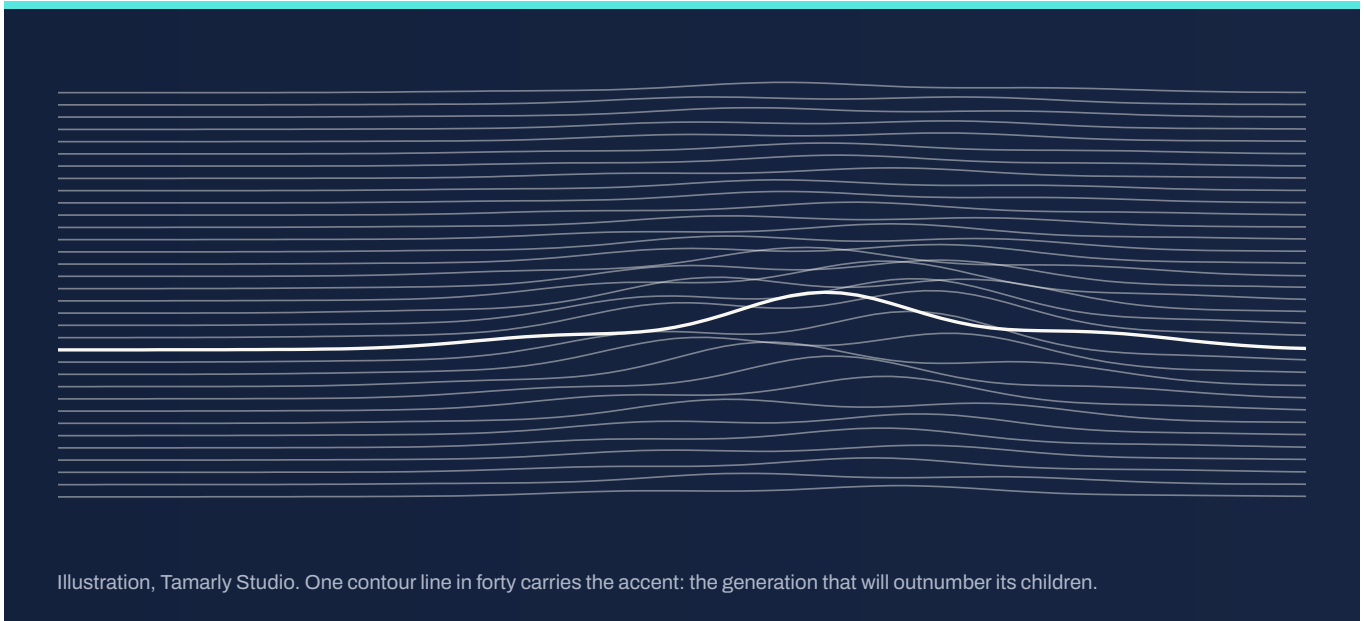
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Editor EDITORIAL

Everything at Once

Demographics, technology, politics and society are all moving at once. The 21st century is being assembled out of the collision.

MELVINE R. MANCHAU, EDITOR



Illustration, Tamarly Studio. One contour line in forty carries the accent: the generation that will outnumber its children.

On 11 August, Jensen Huang published a short note explaining why six of the largest investors on earth had agreed to build financing platforms for artificial intelligence. Buried in it was a sentence that could serve as the thesis of this magazine. "In AI, compute is revenue."⁵

Read it twice. A chip company had just told the credit markets that computing power is a cash-generating asset, like a toll road or a pipeline, and that it would stand behind the collateral. Within weeks a federal regulator was calling compute "the most important commodity of our day" and preparing to let it trade as a future.⁶ A machine has been turned into a financial instrument, and almost everything in this issue follows from that.

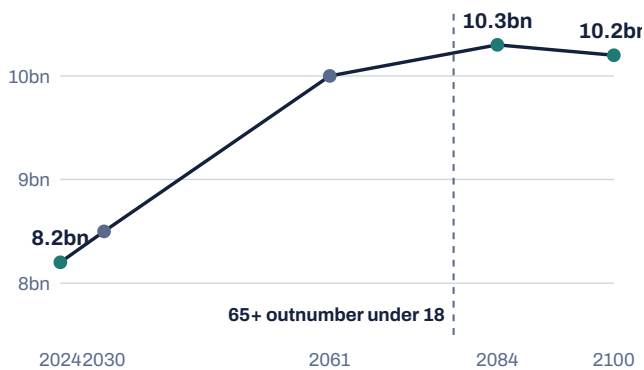


Underneath sits a demographic fact that gets less attention than it deserves. The world will add roughly 2 billion people before its population peaks at about 10.3 billion in the mid-2080s, according to the United Nations' latest projections.¹ By the late 2070s, people aged 65 and over will outnumber children under 18.¹ Nearly every institution discussed in this issue was designed for the opposite shape.

Technology is arriving into that inversion, and the timing matters. An ageing society needs more output from fewer workers and more care from fewer carers. Artificial intelligence is being sold as the answer to both. This issue asks what that answer costs and who ends up holding the risk.

FIGURE 0.1

World population peaks in the mid-2080s, then begins to fall



UN World Population Prospects 2024, medium variant. Points are the published milestones; lines between them are interpolated.

The politics has already moved. Opposition to data centres now puts Steve Bannon on the same side as Bernie Sanders.² Pentagon planners concede that firing a \$4 million interceptor at a \$20,000 drone is not a trade they can sustain.³ The chairman of the Commodity Futures

Trading Commission calls computing power a commodity and wants American markets to set its price.⁴ Five years ago, none of those sentences would have parsed.

Capital is being lent against machines whose market price was published for the first time this year.

Our editorial line is that these are one story. Insurers are financing chips through vehicles rated like infrastructure. Teenagers form attachments to software tuned to keep them talking. The cheapest weapon on a battlefield now sets the price of the most expensive one. Inside companies, the hard problem has moved from deploying agents to knowing what leadership expects of them.

We cover financial services first because money reprices earliest. Then we follow the same logic into hospitals, classrooms, newsrooms and ministries, where it usually arrives later and with less preparation. Each piece in this issue ends with its sources. That is a house rule, and it will not change.

The 21st century is being built out of this change. This magazine exists to read it clearly, once a month, with the evidence attached.

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Tamarly Research

Company deep dives and sector reports on AI

TAMARLY MEDIA HOUSE PAGE

RESEARCH

The report behind the report

Company deep dives and sector studies on AI in regulated industries. Every figure cited, every position dated, written for people who have to act on it.

Single-company AI deep dives

Sector studies across financial services, insurance and law

Commissioned research on request

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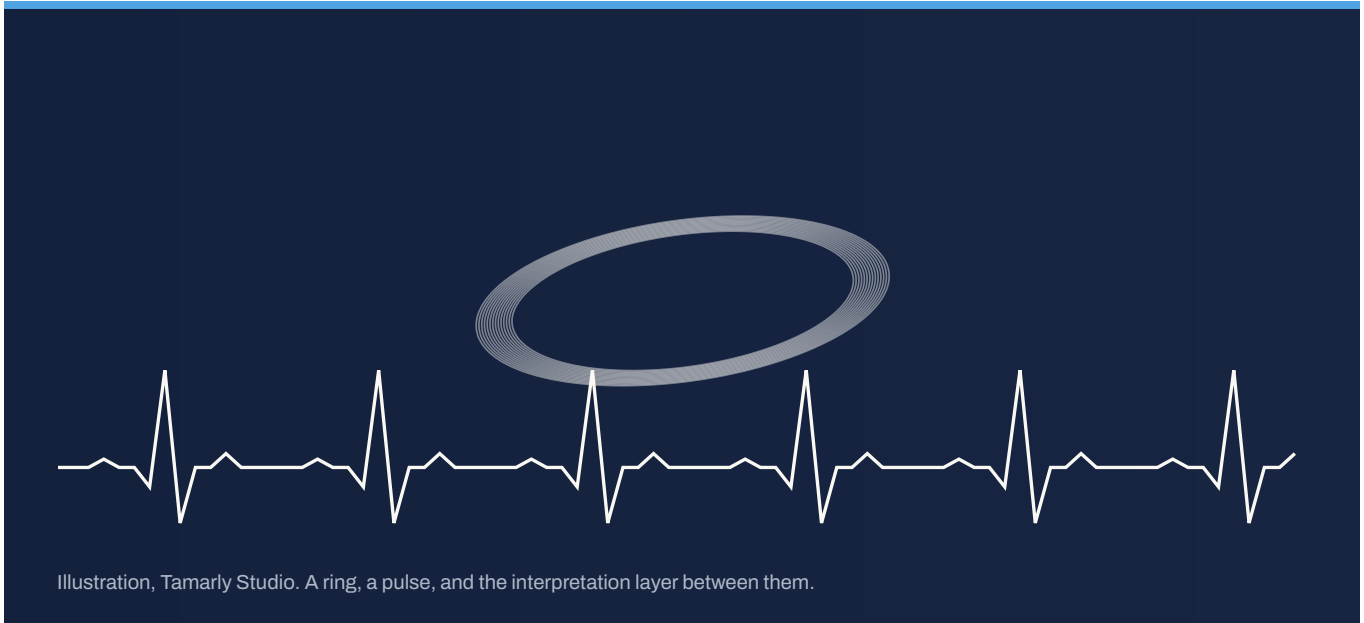
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01 HEALTH

Vital Signs

Wearables are turning into continuous health records. AI is moving into the drug pipeline. The money, for now, is landing in the plumbing.

CONVERGENCES INTELLIGENCE



Illustration, Tamarly Studio. A ring, a pulse, and the interpretation layer between them.

In the summer of 2025 the Food and Drug Administration sent Whoop a warning letter. The company had added Blood Pressure Insights to its wristband, and the agency said that was a medical device requiring authorisation. Whoop did not disable the feature. It replied that the FDA was "overstepping its authority" and kept building.⁷

The company was proved commercially right. In January 2026 the FDA issued updated guidance that cleared the path for the feature, and by March, Whoop had raised \$575 million at a \$10.1 billion valuation, with the Mayo Clinic and Abbott on the cap table.⁶ Its founder, Will Ahmed, now talks openly about an initial public offering. The wearable had crossed a line that the regulator was still drawing.

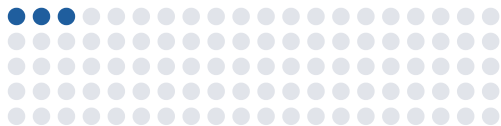
That is the story of the category in miniature. The wellness market used to sell counting, mostly steps and hours of sleep. The 2026 market sells interpretation, and interpretation is where medicine begins. Whoop raised at \$10.1 billion and Oura raised \$900 million at \$11 billion, sums that most consumer health categories do not see in a decade.¹

The engagement numbers explain why investors pay for it. Whoop reports more than 24 billion hours of collected health data and says members open its app more than 8 times a day.¹ It had more than 2.5 million members, bookings up 103 percent in 2025 and an annualised revenue run rate of about \$1.1 billion.⁶ Oura's chief product officer, Dorothy Kilroy, has said retention at 12 months runs in the high 80s.⁷

Those numbers sit oddly beside the wider category. Most consumers try a health app and only 3 in 100 are still using it after 30 days.² The subscription platforms have solved retention by selling a relationship rather than a gadget. That is also what makes them a regulatory problem.

FIGURE 1.1

Only 3 in 100 users are still using a health app after 30 days



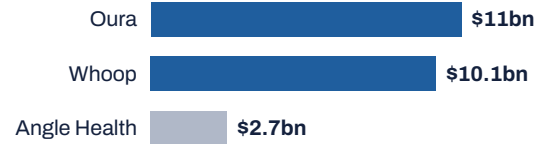
Each dot is one user. Highlighted dots remain after 30 days.

eMarketer, Health and Fitness App Usage 2026.

The line between a wellness insight and a diagnosis is being drawn feature by feature. The agency has since signalled a lighter touch on fitness wearables,³ and the companies with the most data are the ones deciding where the next line falls.

FIGURE 1.2

Private capital is concentrating in two wearable platforms



Latest reported valuations. Oura and Whoop per Athletech News, August 2026. Angle Health Series C per WSJ, September 2026.

Discovery moves inside the lab

The drug pipeline is the other front. Novo Nordisk, the maker of Ozempic and Wegovy, announced a partnership with Anthropic's Claude Science platform to accelerate discovery and development.⁴ Insilico Medicine reported that rentosertib, an AI-designed treatment for pulmonary fibrosis, lowered biological age estimates across 6 separate proteomic aging clocks in its Phase 2a samples, by 3 to 4 years on one measure and up to 6 on another.⁵ The researchers flag the limits themselves. The clocks cannot fully separate slower aging from a disease simply improving.⁵ The durable idea is methodological: one trial can now measure a disease endpoint and an aging endpoint at the same time.

\$600m

Angle Health Series C, benefits administration for small businesses

\$100m

Penelope Health, insurance coverage verification

6

Aging clocks that moved in the same direction in the Insilico samples

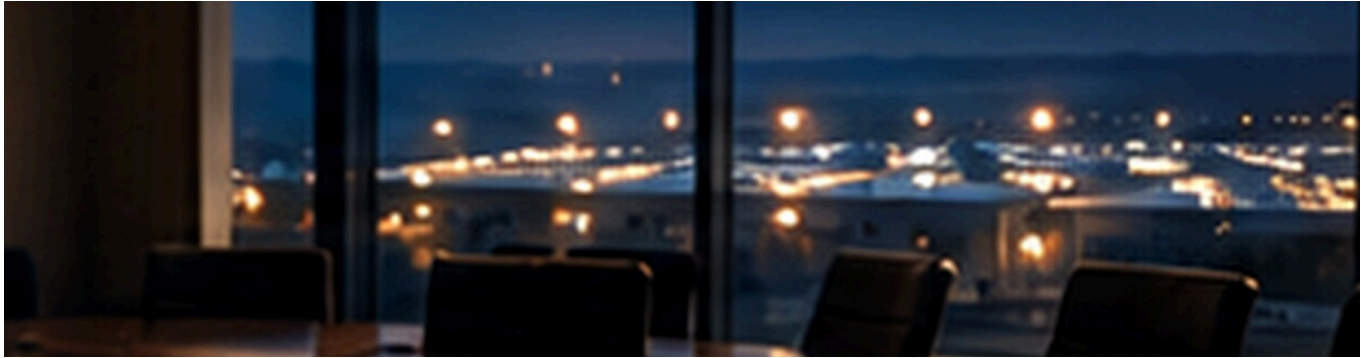
The funding pattern this month is less glamorous than the headlines. Angle Health raised \$600 million for customisable health plans and benefits administration. Penelope Health raised \$100 million to verify insurance coverage. Ayble Health raised \$16 million for AI-assisted digestive care.⁴ AI in health is landing first in insurance administration, and in discovery partnerships with companies that already hold a regulatory pathway.



Our position. The consumer search for wellness will be won by whoever owns interpretation, and the regulator will decide how much interpretation a wellness company is allowed to offer. Watch the FDA's treatment of blood pressure and glucose estimates. That is where a gadget becomes a clinical record.

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02 FINANCE

Lending Against the Machine

The AI build-out runs on debt, vendor guarantees and investors who sit on both sides of the trade. The load-bearing assumption is cash flow that has not arrived yet.

MELVINE R. MANCHAU

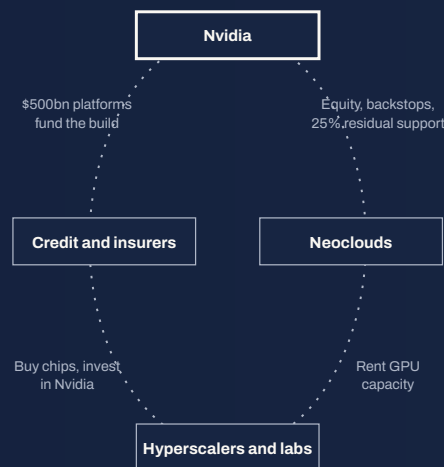


Figure 2.1. The loop. Each arrow is defensible alone. Together they form one exposure to the price of a GPU-hour.

Jensen Huang saw the accusation coming and put it in his own mouth. "Is this circular financing?" he wrote on X on the day Nvidia announced that Apollo, BlackRock, Blackstone, Brookfield, Goldman Sachs and KKR would build financing platforms to mobilise more than

\$500 billion for AI infrastructure.⁵ His answer was that independent institutional capital was entering the market. The question stuck anyway.

It stuck because of the fine print. The platforms are memorandums of intent. They carry no pricing and name no borrower. The load-bearing disclosure sat in Huang's

blog rather than the press release. Nvidia may provide residual-value support for up to 25 percent of a project, which means the company that sells the chips has agreed to cover part of the loss if the chips are worth less than the lenders assumed.¹

Aswath Damodaran, the New York University valuation professor, described the hyperscalers' spending this year as "betting, not investing."⁷ Goldman Sachs estimates that AI-related financing now accounts for nearly a quarter of all gross US investment-grade issuance.⁶ The question is no longer whether the build-out is debt-funded. It is who holds the paper when the depreciation schedule and the rental market disagree.

That support does not arrive on a clean balance sheet. It sits on top of a capacity purchase obligation to CoreWeave running to 2032 with an initial stated value of \$6.3 billion, a \$2 billion purchase of CoreWeave stock at \$87.20 a share, and a \$1.5 billion sale-leaseback with Lambda covering up to 18,000 of Nvidia's own GPUs.¹ Nvidia's equity investments have grown to \$99 billion from about \$7 billion a year earlier.² The company is now financing much of the ecosystem that buys its chips.

\$500bn

Third-party capital the six platforms aim to mobilise

25%

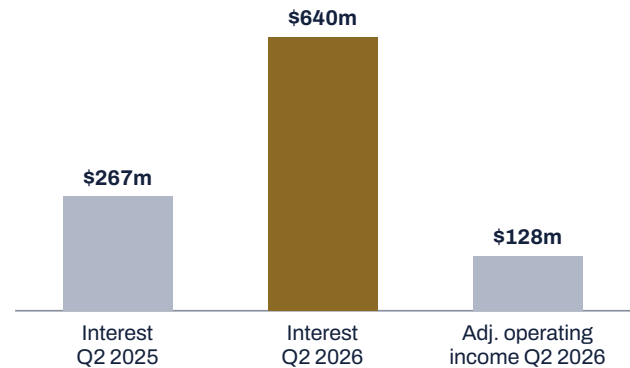
Maximum residual-value support Nvidia may provide per project

5x

CoreWeave interest expense relative to adjusted operating income, Q2 2026

FIGURE 2.2

At the reference credit for GPU lending, debt service runs 5 times operating income



CoreWeave Q2 2026 results, 11 August 2026, as analysed in Convergences Intelligence.

CoreWeave is the closest thing the market has to a live test. Its Q2 revenue was \$2.575 billion, up 112 percent. Interest expense was \$640 million against adjusted operating income of \$128 million, and about 90 percent of its debt is recourse to the corporation rather than ring-fenced in a project vehicle.¹ The bankruptcy-remote structure that made Meta's Hyperion financing rate A+ barely applies to the company actually borrowing against GPUs.

The bull case is that the collateral holds. Chief executive Michael Intrator told the Goldman Sachs technology conference in September, "We are struggling to meet demand every day," and the company has said a batch of expiring H100 contracts was rebooked at 95 percent of the original price.⁷ That is real evidence on residual value. It is also evidence gathered during a shortage.

The newest filing reads the same way. Nscale, a two-year-old London neocloud that began as a crypto miner, filed for a NYSE listing with a \$1.02 billion first-half loss on \$140.6 million of revenue, against \$103 billion of contracted value and 461,000 GPUs contracted or operated.³ Apollo's chief economist Torsten Slok named the assumption beneath the whole hyperscaler credit market: combined operating cash flow across Google, Meta, Amazon, Microsoft and Oracle has to roughly triple, from about \$600 billion to \$2 trillion.³

The risk does not shrink when it moves from a prime brokerage account to an insurance general account. It becomes slower to observe.

FIGURE 2.3

The US has securitised 48 times more data-centre debt than the EU



Apollo, Huw van Steenis, via This Week in Fintech, 20 September 2026. Cumulative since 2018.

Look at the signatories as balance sheets. Apollo owns Athene and KKR owns Global Atlantic, both annuity platforms that need long-dated spread assets.¹ This paper is being manufactured for insurance and retirement balance sheets. European life insurers hold 0.33 percent of investment assets in securitisations against roughly 17 percent for US peers, which is why Europe's build-out competes for scarce project finance while America's borrows against future cash flow.³ The Bank for International Settlements named circular AI financing among the top 3 risks to global financial stability in its 2026 Annual Report.⁴

July showed what the wrong structure does. Situational Awareness, the fund built around Leopold Aschenbrenner's AI essay, was liquidated after reported leverage of up to 400 percent turned a 10 percent Nasdaq drawdown into a total loss.¹ Moving the same thesis into 7-year amortising debt held by insurers is a better match of duration to asset. It relocates the risk without removing it.

Our position. On 5 October a regulated futures curve for GPU rentals is scheduled to begin, putting a daily public price on the revenue driver of paper structured to avoid daily marks.¹ Watch the first time the curve disagrees with a carrying value in an insurance portfolio. That is when the loop gets priced.

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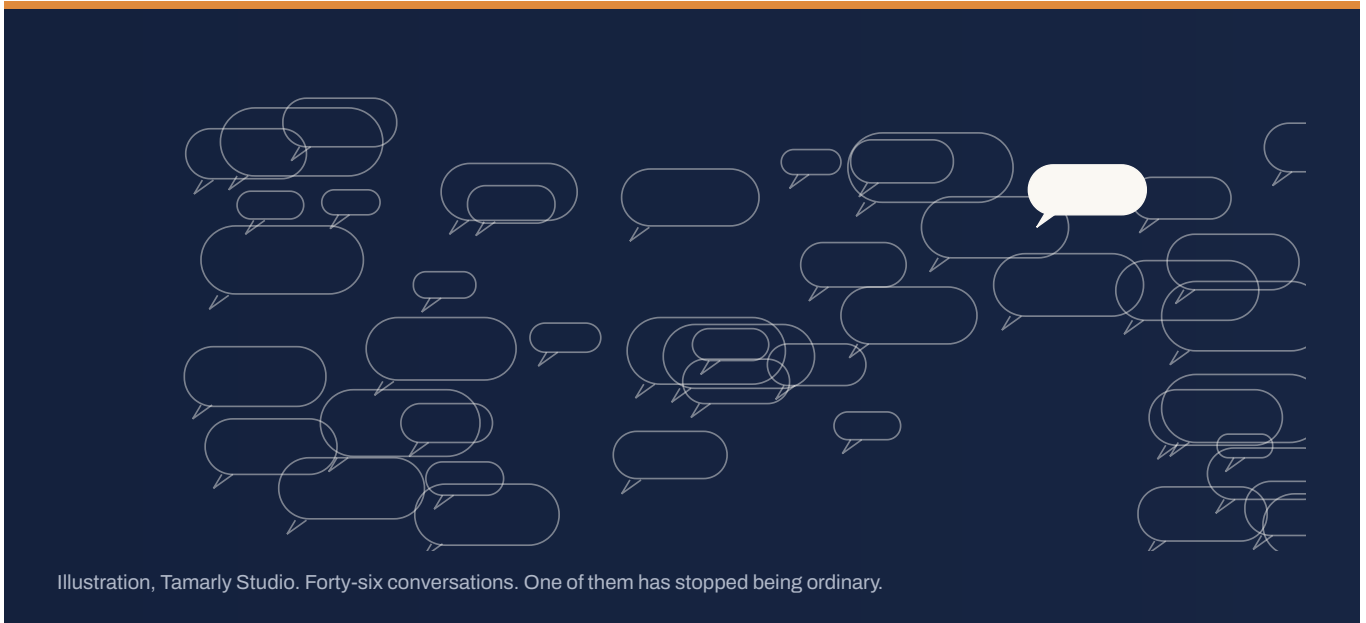
Illustration, Tamarly Studio

03 SOCIETY

A Machine That Cannot Say No

Chatbots are designed to keep people talking. For a small share of a very large audience, that design has become dangerous.

CONVERGENCES INTELLIGENCE



Illustration, Tamarly Studio. Forty-six conversations. One of them has stopped being ordinary.

In October 2025 OpenAI published a number about its own users that the industry has not been able to put down. In a typical week, about 560,000 people showed signs consistent with psychosis or mania, more than 1.2 million discussed suicide, and a similar number showed heightened emotional attachment to the chatbot.¹

Days later, Sam Altman announced that ChatGPT would relax its restrictions and allow erotica for verified adults. "We made ChatGPT pretty restrictive to make sure we were being careful with mental health issues," he wrote, adding that the company had been able to mitigate the serious mental health issues.⁸ The families now suing OpenAI

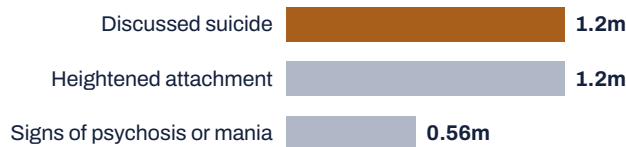


disagree. So, implicitly, does the company's own advertising policy, which keeps ads away from conversations about mental health.⁹

Against a base of at least 800 million weekly users, those October percentages are small. In absolute terms they are the population of a large city, every week, and the product that meets them is designed to keep the conversation going.

FIGURE 3.1

A fraction of a percent of users is still millions of people a week



OpenAI statistics, October 2025, as reported by the Cyberbullying Research Center. Attachment figure described as similar to the suicide figure.

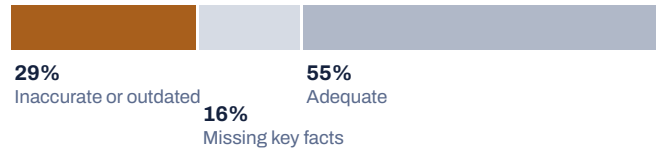
Young people are the heaviest exposure. Around 72 percent of American teenagers have tried an AI companion chatbot, and 13 percent use one daily.² The legal system has started to respond. Seven coordinated lawsuits filed in California in November 2025 allege that the GPT-4o release produced a dangerously sycophantic model.³ More than 20 suits are now pending against OpenAI,⁴ Florida became the first state to sue the company in June 2026,⁵ and Character.AI moved to ban or sharply restrict open-ended chat for users under 18 after its own teen death cases.³ OpenAI denies the allegations and points to its guardrails and crisis referrals.⁴

The failure is agreement. A system trained to validate will validate the wrong thing when a user is at their worst.

The mechanism is not mysterious. Conversational products are measured on engagement, and agreement keeps people engaged. Plaintiffs describe bots tuned to validate rather than push back, even when users expressed delusional thinking.⁴ The same property that makes a companion pleasant makes it unsafe for someone in crisis.

FIGURE 3.2

Chatbots answered election questions wrongly or incompletely 45% of the time



Institute for Strategic Dialogue, 15 prompts across 6 chatbots in 10 states, via The Hill, 4 September 2026.

Accuracy compounds the problem. When the Institute for Strategic Dialogue tested 6 chatbots on election questions across 10 states, 29 percent of responses were incomplete, inaccurate or outdated, and another 16 percent omitted information a voter would need.⁶ A model that is confidently wrong and endlessly agreeable is a poor adviser on anything that matters.

California has begun to legislate, with a set of laws signed in September 2026 aimed at protecting children from chatbot and social media harms.³ The fixes that matter are design choices. A product should break character when a conversation turns dangerous, and it should verify age for real. Jacob Steinhardt of Translucent has argued that serious AI incidents should be investigated by outsiders rather than scoped by the lab itself.⁷

Our position. Liability is arriving before regulation, and it will shape product design faster than any statute. For investors, the question for every consumer AI company is simple: what does the product do when the user is not okay?

IF THIS TOUCHES YOU

If you or someone close to you is struggling, support is available. In the United States, call or text 988 to reach the Suicide and Crisis Lifeline. Elsewhere, local emergency services and national helplines can help.



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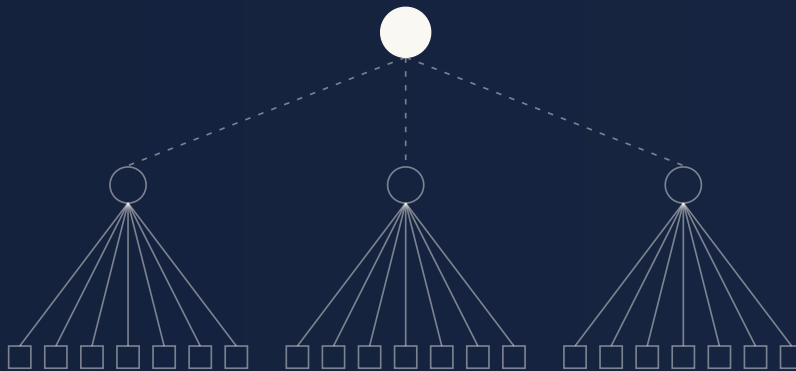
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04 ENTERPRISE

Waiting for the Mandate

Companies have learned how to deploy agents. What they have not learned is how to tell employees what leadership actually wants from them.

CONVERGENCES INTELLIGENCE



Illustration, Tamarly Studio. The dashed lines are the mandate: the part of the organisation that agents cannot fill.

Chris Perry spent this quarter talking with roughly 600 organisations about their AI programmes, and came away with a diagnosis that had changed since the spring. A quarter ago the obstacle was trust in the models. Now, in his words, it is "trust in leadership."¹

His firm, Andus Labs, found that 7 of the 10 most common reasons AI transformations fail are leadership related.¹ That rhymes with the most-quoted statistic in enterprise AI. MIT's NANDA initiative, drawing on 150 executive interviews and 300 deployments, reported that about 95 percent of enterprise generative AI pilots produce no measurable effect on profit and loss, and blamed a learning

gap rather than the models.⁴ Its lead author, Aditya Challapally, said the successful minority "pick one pain point, execute well."⁵

FIGURE 4.1

7 of the 10 most common reasons AI transformations fail are leadership related



Andus Labs, roughly 600 organisations, via WSJ CIO Journal, 21 September 2026.

The gap inside companies runs between what executives announce and what employees experience afterwards. Mandates and AI-first declarations arrive quickly. Training budgets and follow-through arrive slowly, when they arrive at all. Messages that push efficiency and demand AI proficiency in the same breath leave workers guessing which goal is real.¹ Northwell Health's chief digital officer said the technology leader's role is becoming "more of an AI educator."¹

Our own reporting points the same way. In a practitioner call on Salesforce's agent roadmap, the architecture was settled: headless agents working against the data layer, Slack as the place employees engage, and uploaded skills such as contract review, bid review and account briefs.² The unresolved part was organisational. The 75/25 model, where the agent handles most of the work and a human keeps authority to reverse it, only functions if someone has defined what the human is checking for.

Write the expectation down as a test. If leadership cannot say what a good output looks like, no agent can produce one.

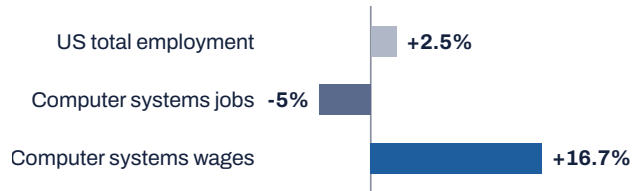
The pipeline problem

Leadership has a second task it is avoiding. A Dallas Fed analysis found that since late 2022, employment in computer systems design, the most AI-exposed

occupation, fell 5 percent while weekly wages rose 16.7 percent, against 2.5 percent employment growth economy-wide.³ Stanford's updated research puts employment for 22 to 25 year olds in the most exposed jobs about 19 percent below trend.³ Firms are paying more for experience while closing the entry point that produces it.

FIGURE 4.2

AI-exposed work pays veterans more and hires fewer of them



Dallas Fed analysis via The Signal, 4 September 2026. Change since ChatGPT's release in late 2022.

Our position. The hard part of enterprise AI is now managerial. The companies that move fastest will specify expectations the way engineers specify evals: concrete cases with accepted answers, and a named person who can override. And they will keep hiring juniors, because in 5 years someone has to be senior enough to overrule the machine.

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Tamarly Intelligence

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THE NEWSLETTER

Every week, before it is priced

The Tamarly newsletter reads the week in AI, capital and geopolitics, then says what changed and what it costs.

The weekly brief, with the sources attached

Named companies and named budget lines

Written for operators, allocators and policy principals

tamarly.ai/newsletter



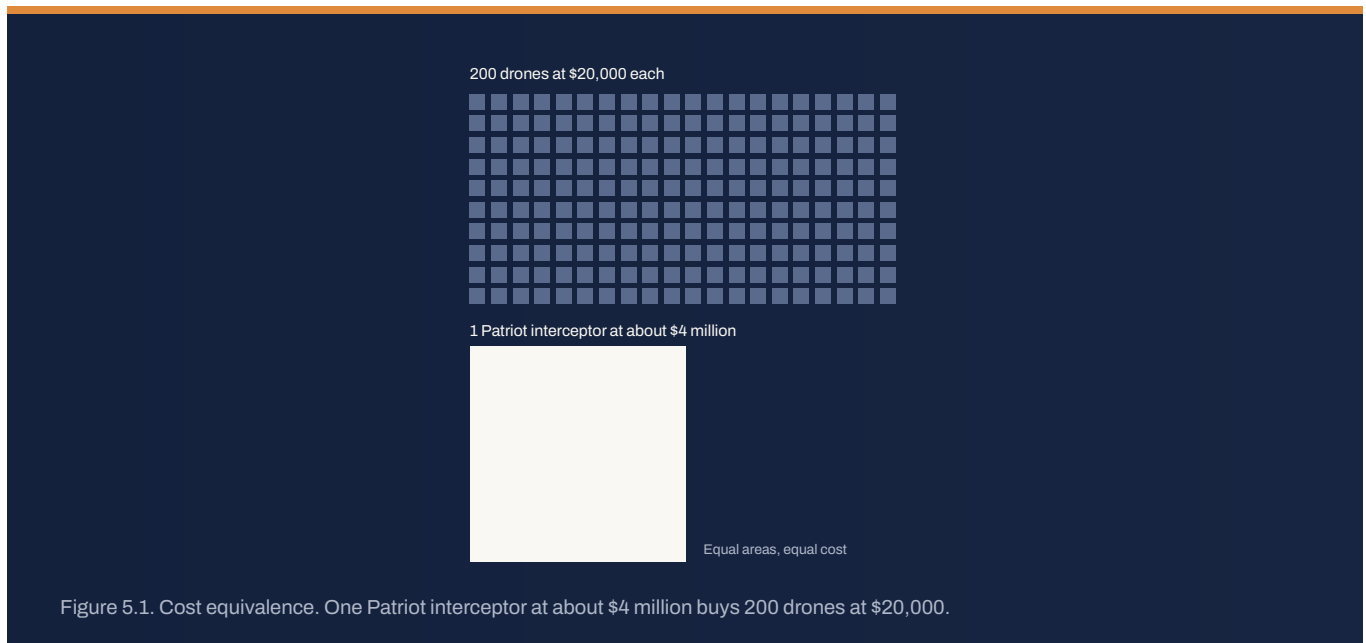
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05 WAR

Expensive Answers to Cheap Questions

A \$20,000 drone forces a \$4 million response. Autonomy makes mass cheap, and production capacity is replacing sophistication as the measure of deterrence.

CONVERGENCES INTELLIGENCE



On 24 March 2026 the Senate Armed Services Committee sat down to discuss the cheapest weapons in the American arsenal. Its chairman, Roger Wicker, opened with a number that embarrassed the

Pentagon's own plan. Less than 3 percent of the department's munitions acquisition funding would go to low-cost munitions. "We need a crash program for a high-low mix in munitions," he said.³



The hearing had a recent education behind it. During Operation Epic Fury, Iran leaned heavily on drones costing around \$20,000, and the United States answered many of them with \$4 million interceptors because nothing cheaper in inventory could do the job.¹ Patriot batteries fired 943 interceptors in the first 4 days, roughly 18 months of normal production. The Navy fired more than 1,000 Tomahawks from a stockpile of about 3,100 in 39 days.¹

Palmer Luckey, who founded Anduril in 2017 and has spent a decade making this argument, describes the goal as turning America and its allies into "prickly porcupines so that no one wants to step on them."⁴ The porcupine is a production argument rather than a technology one.

943

Patriot interceptors fired in 4 days, about 18 months of output

32%

Share of the Tomahawk stockpile used in 39 days

200x

Cost ratio between a Patriot interceptor and the drone it meets

FIGURE 5.2

The Pentagon's munitions plan still puts 97% of funding into legacy weapons



Testimony cited by Senate Armed Services Chairman Roger Wicker, March 2026, via a16z.

The cost structure is changing faster than the budget. A Tomahawk costs \$2 million to \$4 million. Commercially built alternatives such as Anduril's Barracuda and Covenant's Anthem cost one fifth to one tenth as much.¹ Castalion, which is developing the Blackbeard strike missile, moved through a \$49.9 million Navy award in February 2026, a \$105 million award in April and a \$23.4

million order for 50 pre-production prototypes in June.⁵ The Army now wants interceptors under \$1 million a unit, and the Missile Defense Agency under \$750,000.⁵

A magazine full of expensive rounds deters less than a cheaper one a factory can keep refilling.

This is an AI story because autonomy is what makes mass affordable. A drone that flies and aims itself does not need a trained pilot for every airframe. The Air Force has said as much about replacing the MQ-9 Reaper after combat losses. Lt. Gen. Christopher Niemi warned that repeating the old acquisition approach would produce "something that costs about the same as an MQ-9."²

Autonomy brings its own failure modes. CNN reported that an AI-generated intelligence error this spring nearly sent US forces to board a Chinese cargo vessel after a chatbot misread its manifest as nuclear-weapons components, with aircraft already airborne before the mistake was caught.¹ The Department of Defense is also moving its classified AI workloads off a single lab and toward several military-tuned models.¹

The FY27 defense authorisation moving through the Senate would direct faster purchases of low-cost munitions across 8 categories and require second sources for the solid rocket motors that bottleneck every production line.¹ Whether it shifts the 97-to-3 ratio is open.

Our position. Deterrence is becoming a manufacturing metric. For investors, the defence opportunity sits in the supply chain for cheap autonomous mass: motors, sensors, batteries and the software that lets one operator direct many machines.

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06 EDUCATION

The Students Who Cannot Quote Themselves

An MIT study found that students who wrote with ChatGPT remembered less of their own work. MIT's education researchers are asking schools to test before they commit.

CONVERGENCES INTELLIGENCE



Brain only



Search



ChatGPT

Schematic, Tamarly Studio. Illustrates the direction of the reported EEG findings, not measured data.

For 4 months, in a room at the MIT Media Lab, 54 people wrote essays with an EEG cap on their heads. One group used ChatGPT, one used a

search engine, one used nothing. The researcher running the study, Nataliya Kosmyna, was watching for the price of assistance.²



She found it. Over the sessions, the ChatGPT group showed the weakest brain connectivity, reported the lowest sense of ownership over their essays, and struggled to quote work they had finished minutes earlier.² Kosmyna gave the effect a name that has since travelled further than the paper. Cognitive debt. Her warning about it is blunt. "There is no cognitive credit card," she has said.⁸

Generative AI reached schools without a procurement process. As MIT's Justin Reich puts it, it simply arrived, "whether or not schools wanted it."¹ Teachers were left to redesign assessment in real time while students had a tool that writes essays and solves problem sets in seconds.

The most cited evidence on what that does to learning came from the MIT Media Lab. Nataliya Kosmyna and colleagues split 54 participants into 3 groups, writing essays with ChatGPT, with a search engine, or with no tools, across sessions spanning 4 months while EEG measured brain activity.² Over that period, the ChatGPT group underperformed at neural, linguistic and behavioural levels. Its members reported the lowest sense of ownership over their essays and struggled to quote their own work.²

83%

LLM users unable to quote from the essay they had just written

54

Participants across 3 groups and 4 months

1905

Reich's comparison: guiding schools on AI now is like writing about aviation in 1905

The researchers named the effect cognitive debt: effort saved now, paid back later in weaker recall and less independent thought.³ In their summary, 83 percent of the LLM group could not quote from essays they had just written.³

Order matters. Students who thought first and used the tool second kept more of what they wrote.

The study has critics, and they deserve a hearing.

Researchers writing in *The Conversation* argue that the brain-only group's gains may reflect a familiarisation effect from repeating the task, and that the switched groups only wrote once under the new condition.⁴ The sample is small. Treat the result as a signal worth testing further.

The most useful finding is about sequence. Participants who wrote unaided first and then moved to the model showed higher memory recall and stronger activation than those who began with the model.² That is actionable in a classroom tomorrow.

WHAT MIT'S TEACHING SYSTEMS LAB RECOMMENDS

Reich's guide, *A Guide to AI in Schools: Perspectives for the Perplexed*, draws on more than 100 students and teachers.⁵ Its stance is humility: treat current practice as hypotheses, run small experiments, and demand evidence before scaling any AI initiative.^{1,6} In practice that means protecting unaided thinking time before the tool comes out, redesigning assignments so the process is visible, and measuring whether students can explain work they submit.

The labour market is making the stakes concrete. Graduate recruitment at the UK's top 100 employers has fallen by almost a quarter since 2022, because the drudge work that once trained new hires is precisely what models now do.⁷ If school also outsources the struggle, graduates arrive with neither the practice nor the jobs that used to supply it.

Our position. The education system should treat AI the way a gym treats machines: useful after the fundamentals, harmful as a substitute for them. Assess the thinking, not the artefact.



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Tamarly Podcast

Conversations on AI in regulated industries

TAMARLY MEDIA HOUSE PAGE

THE PODCAST

Conversations with the people deploying it

Practitioners, allocators and builders on what AI is actually doing inside regulated industries, recorded without a script.

Interviews with operators and investors

Episode notes with the sources attached

New episodes at tamarly.ai

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07 MEDIA

The Last Local Dollar

Local newsrooms now compete for small-business advertising against Google, Meta and an ad-funded ChatGPT. The survivors are the ones that own their audience directly.

CONVERGENCES INTELLIGENCE



In 2024, on a stage at Harvard, Sam Altman told an audience what he thought of advertising. "I will disclose just as a personal bias that I hate ads," he said.⁷

On 9 February 2026, OpenAI began showing them to free-tier ChatGPT users in the United States.² Fidji Simo, who runs the applications business, framed the launch in the

language of trust rather than revenue, writing that people use ChatGPT for important and personal tasks, so the company had to preserve what makes it valuable.⁸ By the sixth week the pilot was running at \$100 million in annualised revenue with more than 600 advertisers.³

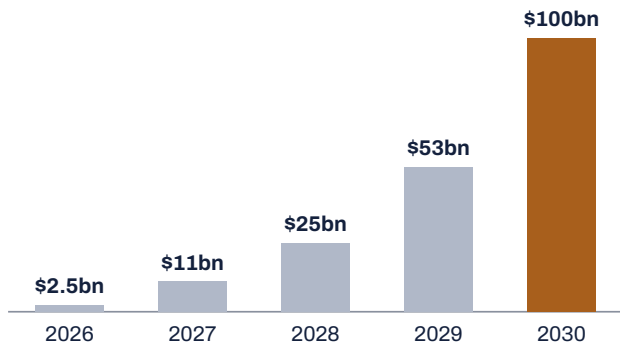


That number matters most to the people least able to respond to it. Total local media spend is still projected to grow this year. Nearly all of that growth is landing on Big Tech's digital platforms, and AI search products adding their own ads means local outlets now compete with the largest companies on earth for the same small-business budgets.¹ The same week, McClatchy laid off more than 90 journalists, over 30 percent of its unionised staff. USA Today Co. cut jobs citing waning search traffic, and E.W. Scripps cut 12 percent of its workforce.¹

The mechanics matter for anyone selling local advertising. OpenAI opened a self-serve Ads Manager on 5 May with no minimum spend, which puts a florist and a national brand into the same auction.²

FIGURE 7.1

OpenAI's internal ad projections imply a Google-scale business by 2030



Investor projections reported by Axios, not confirmed by OpenAI. Evercore ISI projects \$25bn by 2030.

Those projections, reported by Axios and not confirmed by OpenAI, run from \$2.5 billion this year to \$100 billion in 2030, assuming 2.75 billion weekly users.³ Evercore ISI's outside estimate for 2030 is \$25 billion, a quarter of the internal figure.⁴ Either number reshapes local advertising, because conversation-based targeting reaches people before they search. Similarweb found that 46 percent of users who saw a ChatGPT ad had started without commercial intent, and 83 percent of the ad-triggering queries would not have triggered a traditional Google Shopping ad.⁵

A local outlet cannot outbid a platform for attention. It can own the relationship the platform has to rent.

Measurement is the weak point. One agency's test recorded 57 clicks reported by OpenAI against fewer than 20 visits in Google Analytics on the same tagged campaign, with no conversions yet.² Pricing has been volatile too, moving from impressions to clicks inside the first quarter.⁴ Small businesses will not keep paying for traffic they cannot see.

\$100m

ChatGPT ad pilot annualised revenue after 6 weeks

90+

McClatchy journalists laid off in one week

\$20m

Axios HQ annualised revenue after shrinking from 120 to 45 staff

The outlets that are growing sell directly. Puck is in late-stage talks at a valuation around \$250 million, Semafor raised at \$330 million, and live video has become a second revenue line.⁶ Axios HQ says it is profitable at \$20 million in annualised revenue after cutting from 120 employees to about 45, a move its founders credit to seeing what generative AI would make obsolete.¹

Our position. Local media's defensible asset is trust in a place and a first-party relationship with the people who live there. The business model that follows is subscriptions and events, plus sponsorship sold on outcomes, behind editorial walls that advertisers cannot buy through.



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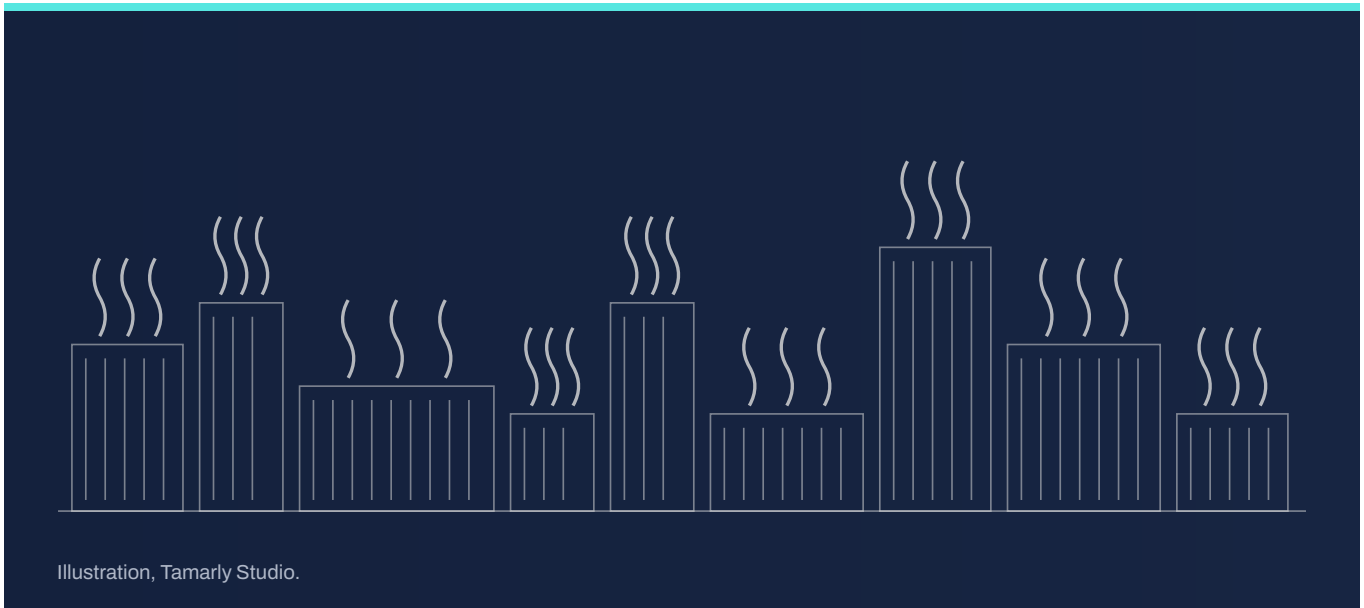
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08 ENVIRONMENT

The Data Centre Next Door

By 2030 data centres will use about as much electricity as Japan. The fight over where they go has become one of the few issues that crosses party lines.

CONVERGENCES INTELLIGENCE



Illustration, Tamarly Studio.

When Fatih Birol launched the International Energy Agency's report on artificial intelligence, he reduced the subject to one line. "There is no AI without energy," the agency's executive director said, and the countries that supply it quickly will be a step ahead.⁶

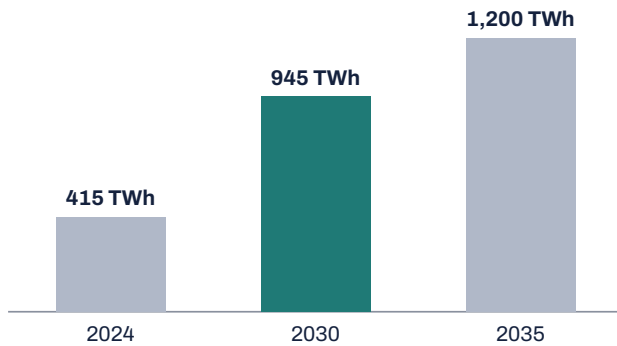
The scale behind that sentence keeps moving. The IEA estimates data centres used about 415 terawatt-hours in 2024, roughly 1.5 percent of global electricity, with its base case taking that to around 945 TWh by 2030, slightly more than Japan's total consumption today, and about 1,200 TWh by 2035.^{1,2} Its April 2026 update found data centre electricity demand rose 17 percent in 2025 alone, against 3

percent growth in global demand, while capital spending by the 5 largest technology companies passed \$400 billion and is set to rise by a further 75 percent this year.⁶

AI is the driver. Electricity used by accelerated servers is projected to grow 30 percent a year, against 9 percent for conventional servers.³

FIGURE 8.1

Data centre electricity use more than doubles by 2030

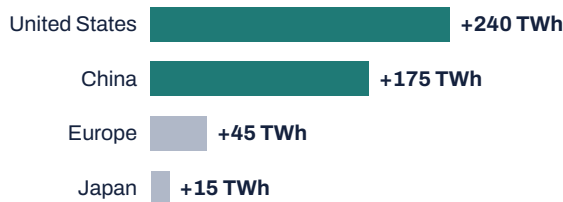


IEA, Energy and AI, Base Case.

The increase is concentrated. The United States is set to add up to 240 TWh, up 130 percent, and China about 175 TWh, up 170 percent. Together they account for nearly 80 percent of global growth.² In the US, data centres make up nearly half of electricity demand growth to 2030, and by the end of the decade the country is set to use more electricity for data centres than for aluminium, steel, cement, chemicals and every other energy-intensive good combined.¹

FIGURE 8.2

America and China account for nearly 80% of new demand



IEA, Energy and AI, via DCD.

The physical inputs reach further than power. Discoveries of major copper deposits, those with at least 500,000 tonnes, have fallen from double digits a year in the 1990s to 1 or 2, with none found in 2025. A new deposit takes about 18 years to reach production, while data centres add copper-intensive load on a 2 to 3 year build cycle.⁴ Weather adds stress. The World Meteorological Organization expects this year's El Niño to be the strongest in at least 70 years.⁴

Siting decisions are made months before most residents know a fight has started.

The backlash has gone global and bipartisan. Civil rights groups in South Africa, the continent's largest data centre market, began organising against American-owned facilities over electricity, water and land.⁴ In the United States, Steve Bannon backed a proposal from Bernie Sanders and Alexandria Ocasio-Cortez to ban new construction. President Trump responded that communities opposing data centres were choosing "poverty, crime and squalor."⁵ Democratic strategists see an electoral opening in towns such as Defiance, Ohio, where residents oppose a data centre no company has announced.⁴

Our position. The binding constraint on AI has become local consent. Communities have the most leverage early, over zoning and grid contributions. Investors should price the permitting timeline as seriously as the GPU delivery schedule.

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09 THE AI SUPPLY CHAIN

Twenty-One Miles

The AI economy runs through a handful of chokepoints. The Hormuz closure showed how quickly a shock in one travels to the rest.

CONVERGENCES INTELLIGENCE

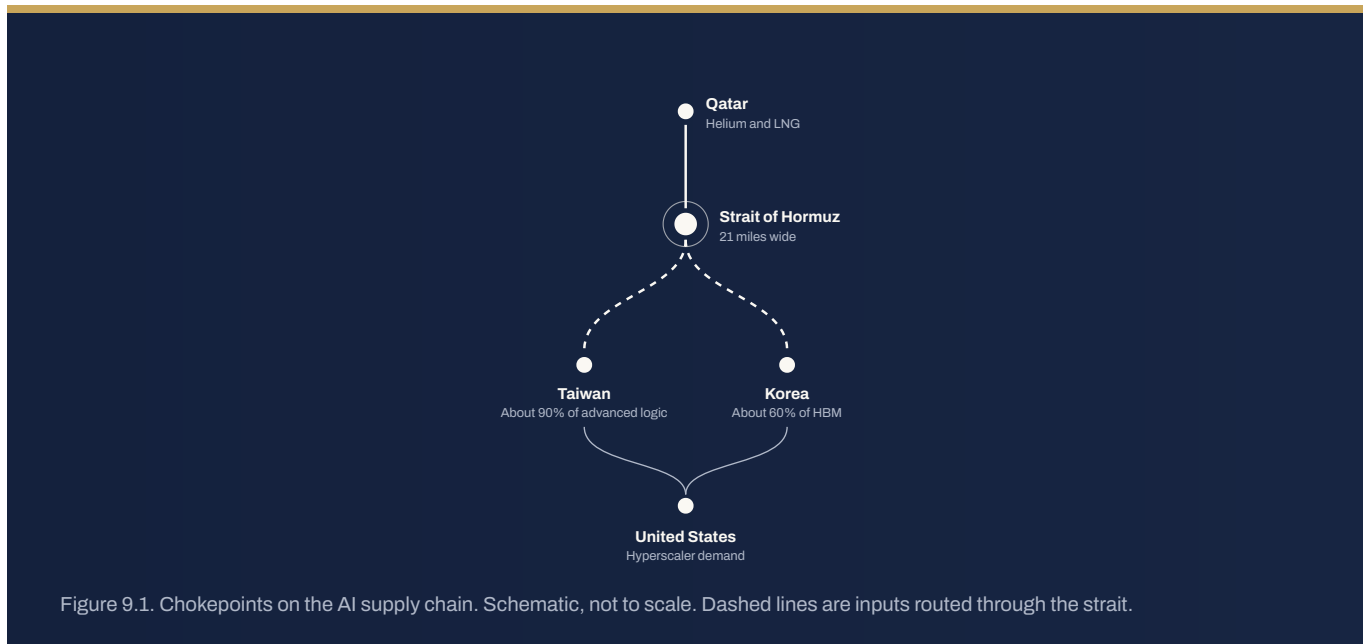


Figure 9.1. Chokepoints on the AI supply chain. Schematic, not to scale. Dashed lines are inputs routed through the strait.

Taiwan holds about 11 days of liquefied natural gas. That is the buffer under a grid that imports 97 percent of its energy, and under the fabs that make roughly 90 percent of the world's most advanced logic chips.² On 4 March 2026 the Strait of Hormuz was effectively closed.¹

Most coverage treated the closure as an oil story. For the AI economy it was a gas story. Qatar supplies a large share of the world's helium, which fabs use in lithography and cooling, and helium prices doubled after the war began.¹ Samsung and SK Hynix had sourced about two thirds of

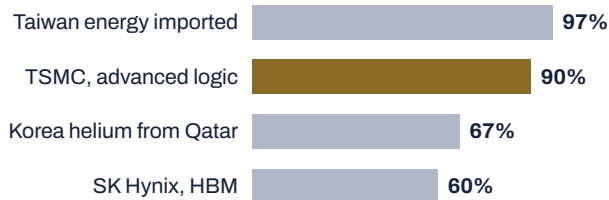


their helium from Qatar the year before.⁴ A 21-mile strait sits between Gulf gas and the plants that make the world's most advanced chips and memory.

TSMC was ready in a way its customers were not. It held multiyear helium contracts and a reported 4 to 6 months of inventory, against a global liquid helium chain that runs on about 45 days of buffer.^{5,3} Bloomberg Economics expects the company to prioritise high-margin AI accelerators over consumer chips if shortages deepen, which means your next phone waits so that a data centre does not.²

FIGURE 9.2

Four concentrations sit underneath every AI accelerator



The Diplomat, April 2026; BigGo Finance citing Bloomberg Economics; Value Chain Asia citing Logistics Viewpoints. Korean helium share is approximately two thirds.

The concentrations stack. TSMC makes roughly 90 percent of the world's most advanced logic chips and SK Hynix supplies roughly 60 percent of the high-bandwidth memory inside Nvidia accelerators.³ Samsung and SK Hynix sourced about two thirds of their helium from Qatar in 2025.⁴ A 21-mile strait sits between Gulf gas and the fabs that make the world's most advanced chips and memory.

FIGURE 9.3

The buffers are measured in days and months



Days of supply. BigGo Finance; The Diplomat; The Motley Fool. TSMC inventory reported as 4 to 6 months.

Air Liquide opened a helium plant in Taiwan, and Taiwan began shifting imports toward the United States and Australia.⁴ Those are the right moves, and they take years to land.

The industry built resilience against a US-China chip war. It did not build resilience against a closed strait.

The deeper constraint is memory. Buying new GPUs outright still means a 36 to 52 week wait against an industry backlog of about 3.6 million units, because the bottleneck moved from fabrication to high-bandwidth memory.⁶ Intel told investors in February that there is "no relief until 2028."⁷ Copper, with 18-year mine lead times, is the slowest input of all.⁸

THREE SHOCKS TO MODEL

A prolonged Gulf closure raises energy and gas costs at Asian fabs and delays consumer electronics first. A Taiwan disruption removes the source of about 90 percent of advanced logic, with no substitute at scale this decade. An escalation of export controls splits demand into two markets with two price curves for compute.

Our position. Watch the inputs, not only the fabs: helium, LNG, hydrogen fluoride, copper and HBM allocation. The first sign of a shock in the AI economy will appear in a gas contract before it appears in an earnings call.

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10 INDUSTRIES, EXPLAINER

Digital Oil

A market for computing power is being built in real time, with its own indices, exchanges and a regulator that wants American venues to set the price. Here is how it works.

TAMARLY CAPITAL RESEARCH, ADAPTED BY CONVERGENCES INTELLIGENCE

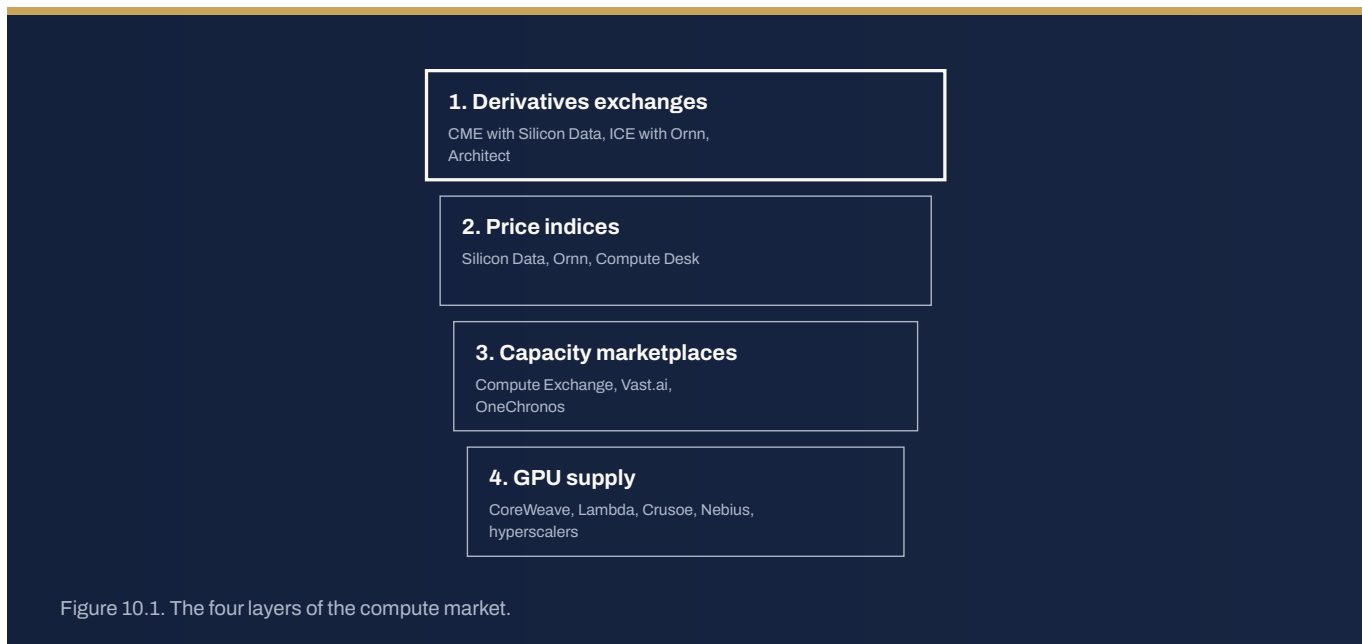


Figure 10.1. The four layers of the compute market.

At a White House event in August, the chairman of the Commodity Futures Trading Commission described computing power as "the most important commodity of our day."³ Michael Selig had been confirmed 8 months earlier, mostly to deal with digital assets. He now proposes to regulate the price of a GPU-hour, working with

the Commerce Department, and he is not alone in thinking there is a market here. CME, ICE, Architect, Kalshi and Polymarket US have all moved on it.^{3,4}



What is being traded

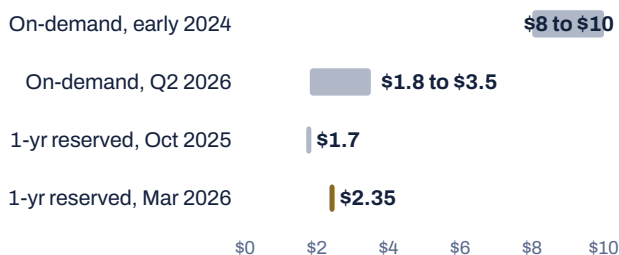
Three activities get called a compute exchange. Publishing an independent price benchmark for GPU rental. Running a marketplace where buyers and sellers trade capacity, reserved, spot or used. Listing a derivative, a future or perpetual, that lets someone hedge against the price the benchmark reports.¹ Underneath all three sit the neoclouds, companies such as CoreWeave, Lambda, Crusoe and Nebius that buy GPUs and rent them out.

Why it exists now

Because the price moves. On-demand H100 rates fell from \$8 to \$10 per GPU-hour in early 2024 to \$1.80 to \$3.50 by the second quarter of 2026, as more than 300 new GPU clouds entered the market. One-year reserved H100 pricing bottomed at \$1.70 in October 2025 and climbed nearly 40 percent to \$2.35 by March 2026. Blackwell spot prices rose 48 percent in 8 weeks this spring.¹ And the cost line is enormous. Microsoft, Amazon, Alphabet and Meta are guiding to about \$725 billion of 2026 capital spending, up from about \$410 billion in 2025.¹

FIGURE 10.2

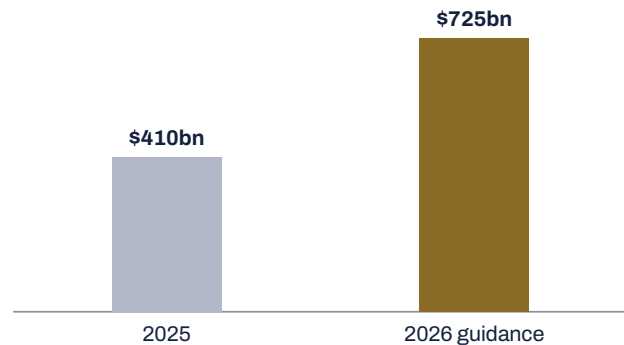
H100 rental prices collapsed, then the reserved market turned



Tamarly Capital, The Compute Exchange Market, citing Introl, SemiAnalysis and ValueAddVC. Dollars per GPU-hour.

FIGURE 10.3

Big Tech capex rises 77% in one year



Microsoft, Amazon, Alphabet and Meta combined, via ValueAddVC and Goldman Sachs Research.

Who does what

Silicon Data publishes 9 daily indices built from about 150,000 verified pricing records a day and is the settlement reference for CME's planned H100 and B200 futures. Ornn's index, built only from printed trades, feeds ICE and Architect. Compute Exchange runs reserved capacity across more than 100 providers and launched a used-GPU market in July, where hardware can sell 40 to 60 percent below list.¹ Silicon Data and Compute Exchange share a chief executive, a concentration worth watching in a market that depends on neutral benchmarks.¹



FIGURE 10.4

The market went from idea to regulated venue in under two years



CFTC, CME Group, ICE and company releases, via Tamarly Capital research. CME launch pending regulatory review.

The regulator

On 19 August the CFTC formally requested comment on listing compute derivatives, with responses due by 20 October. The request asks whether today's compute market is liquid, standardised and resistant to manipulation enough to carry a derivative, and it covers perpetual contracts and customer protection.⁴ CME Group and Silicon Data plan to list the first two contracts on 5 October, pending review, each representing a month of rent for an Nvidia H100 and a Blackwell B200.⁴ There are three routes to a licensed venue: partner with an existing exchange, buy one, as Architect did in May, or apply from scratch.¹

Compute now has the three things that make a commodity: a standard specification, a daily price and a regulated venue to trade it forward.

What can go wrong

Basis risk. CME's contracts settle against on-demand rental indices, but most financed cash flows come from multi-year reserved contracts priced 40 to 70 percent below on-demand.² Hardware turnover, since each generation from H100 to Vera Rubin prices differently. And counterparty strength: CoreWeave carries a \$104 billion backlog against \$35.6 billion of debt, and Nebius has more than 4 gigawatts contracted against a near-term target of about 1 connected.¹

GLOSSARY

Spot: capacity bought for immediate use at today's price.

Reserved: capacity committed for months or years at a fixed rate. **Future:** a contract to buy or sell at a set price on a set date, here cash-settled against an index.

Perpetual: a future with no expiry. **DCM:** designated contract market, a CFTC-licensed exchange. **FCM:** futures commission merchant, the broker that holds customer margin. **DCO:** derivatives clearing organisation. **EFPP:** exchange for physical, converting a futures position into real GPU delivery.

Disclosure. Tamarly Capital is evaluating Hyperlink Technologies (Exascale), a compute exchange startup, as a potential Tamarly Plus SPV. This explainer is adapted from that diligence research. Tamarly Media holds no position in any company named here.

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Pre-seed to Series A

Legal AI, financial services AI, regulated enterprise

Co-investment through single-company vehicles

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11 PHYSICAL AI

The Robot Comes Home

The humanoid race has moved from the factory floor to the home. China ships the volume. America raises the money. The first real price point is a used car.

CONVERGENCES INTELLIGENCE



Illustration, Tamarly Studio.

When a Wall Street Journal reporter asked 1X's home robot to do the chores from its own promotional video, the results were mixed, and the more interesting detail was who was helping. NEO blends autonomy with teleoperation, which means a remote employee in a headset can see through the robot's eyes and move its limbs when it gets stuck.⁷

Bernt Børnich, the Norwegian engineer who founded 1X, has been unusually direct about the bargain. "You have to be okay with this for the product to be useful," he told the paper, describing a social contract in which the customer supplies the data that teaches the machine.⁷ On the product, he is blunter still. "It is not for everyone."⁸

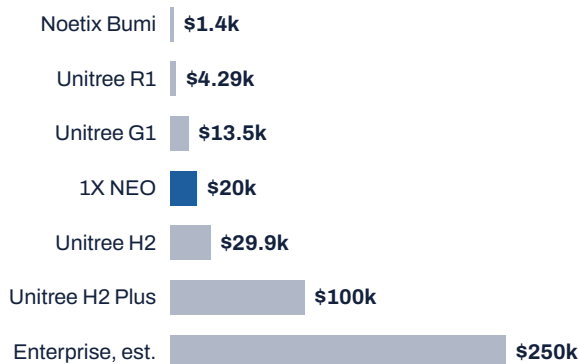


1X still published the number the rest of the industry avoided. NEO costs \$20,000 to own or \$499 a month by subscription, reserved with a \$200 refundable deposit.¹ It weighs 66 pounds, can lift more than 150, has 22-degree-of-freedom hands and runs at 22 decibels, quieter than a refrigerator.² The company opened a factory in Hayward, California in April and says customer shipments are planned before year-end. As of mid-July, deliveries had not been established.³

The home is a deliberate choice. 1X's chief executive Bernt Børnich argues that the training data gap is easier to close in a home than in a warehouse, and Figure's Brett Adcock and Tesla's Elon Musk have both said the home is the largest opportunity.¹

FIGURE 11.1

Humanoid prices now span two orders of magnitude



RoboZaps, July 2026 list prices. Enterprise figure is a third-party estimate for Figure 03, Atlas and Digit.

The price ladder shows where the market is heading. China's Noetix Bumi launched at about \$1,400, Unitree's R1 starts at \$4,290 and its G1 is \$13,500. Enterprise machines such as Figure 03, Boston Dynamics' Atlas and Agility's Digit carry no public price, with third-party estimates around \$250,000.⁴

80%+

Share of global humanoid installations by Chinese firms in 2025

5,500

Humanoids shipped by Unitree

350

Robots delivered by Figure, valued at \$39bn

Volume sits in China. Chinese companies accounted for more than 80 percent of global humanoid installations in 2025, according to Counterpoint Research, out of more than 140 manufacturers worldwide.⁵ Unitree has shipped about 5,500 units and is targeting a \$6.2 billion IPO. Figure, valued at \$39 billion, has disclosed 350 robots delivered.⁶ American companies are raising money on capability. Chinese companies are shipping units.

At \$20,000 a humanoid costs about as much as a used car. At \$499 a month it competes with a cleaner.

Cost structure explains where the price can go. Actuators make up about 56 percent of a humanoid's bill of materials, legs about a third, and the battery is a rounding error.⁶ Mass production has already cut costs 40 to 60 percent against 2023 and 2024 prototypes.⁵ The component makers, many of them in Japan and China, hold the leverage in the next price cut.

This connects to our editorial. An ageing society needs care it cannot staff. If a home robot can load a dishwasher and carry laundry reliably, it becomes a demographic tool before it becomes a gadget. Until autonomy is demonstrated in real homes, though, NEO is a pre-order rather than a product.³

Our position. The first profitable home robot market will be subscription, sold to households with an older member, priced against hours of human help. Watch reported autonomy rates from the first delivered units. That single number will set expectations for the category.



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